FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

INTRODUCTION

Project Overview :

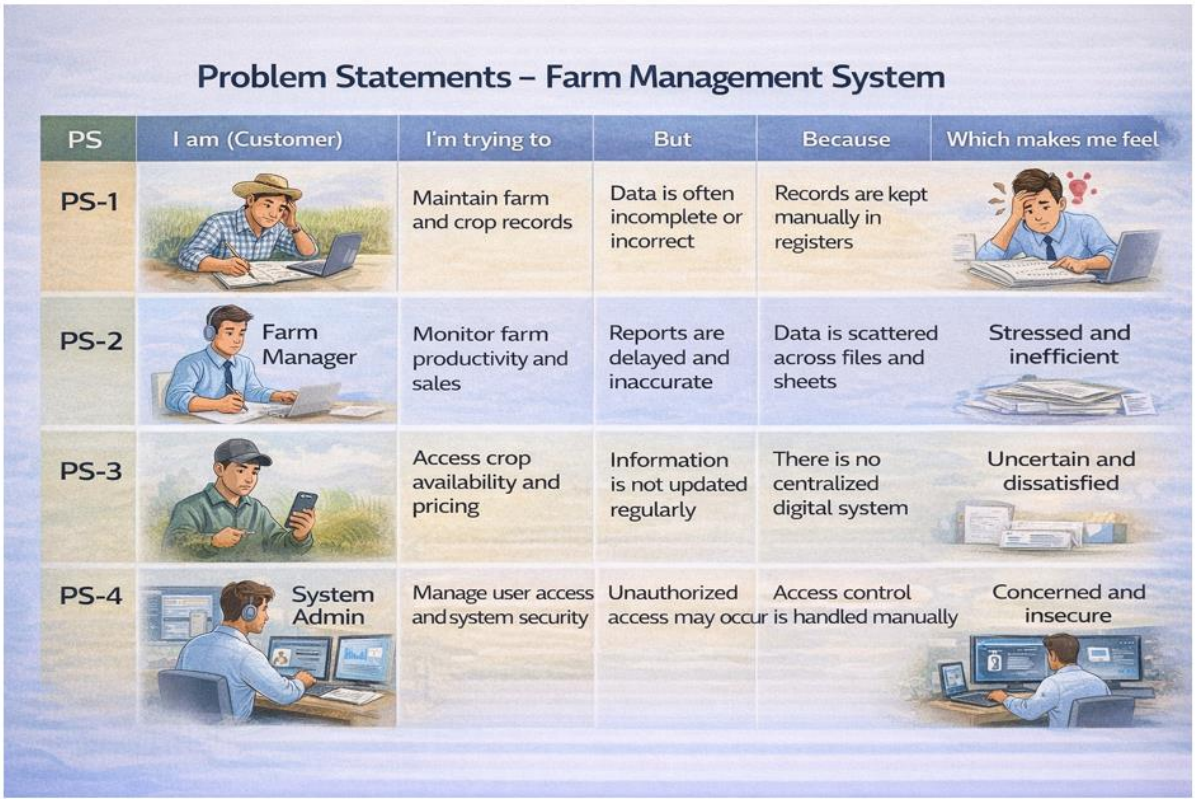
The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

Purpose :

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

IDEATION PHASE

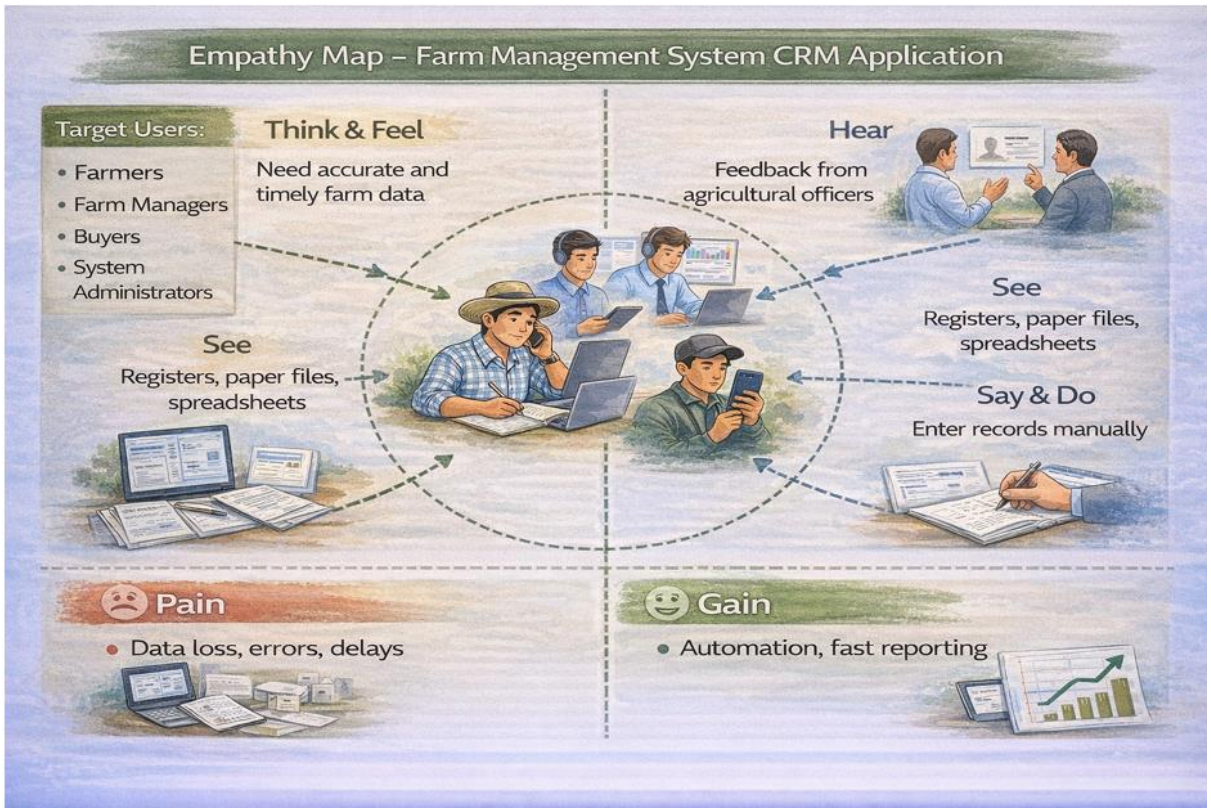


Problem Statements – Farm Management System

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3		Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

Empathy Map Canvas Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators Empathy Analysis
- Think & Feel: Need accurate and timely farm data
- Hear: Feedback from agricultural officers
- See: Registers, paper files, spreadsheets
- Say & Do: Enter records manually
- Pain: Data loss, errors, delays
- Gain: Automation, fast reporting



Brainstorming Brainstormed Ideas

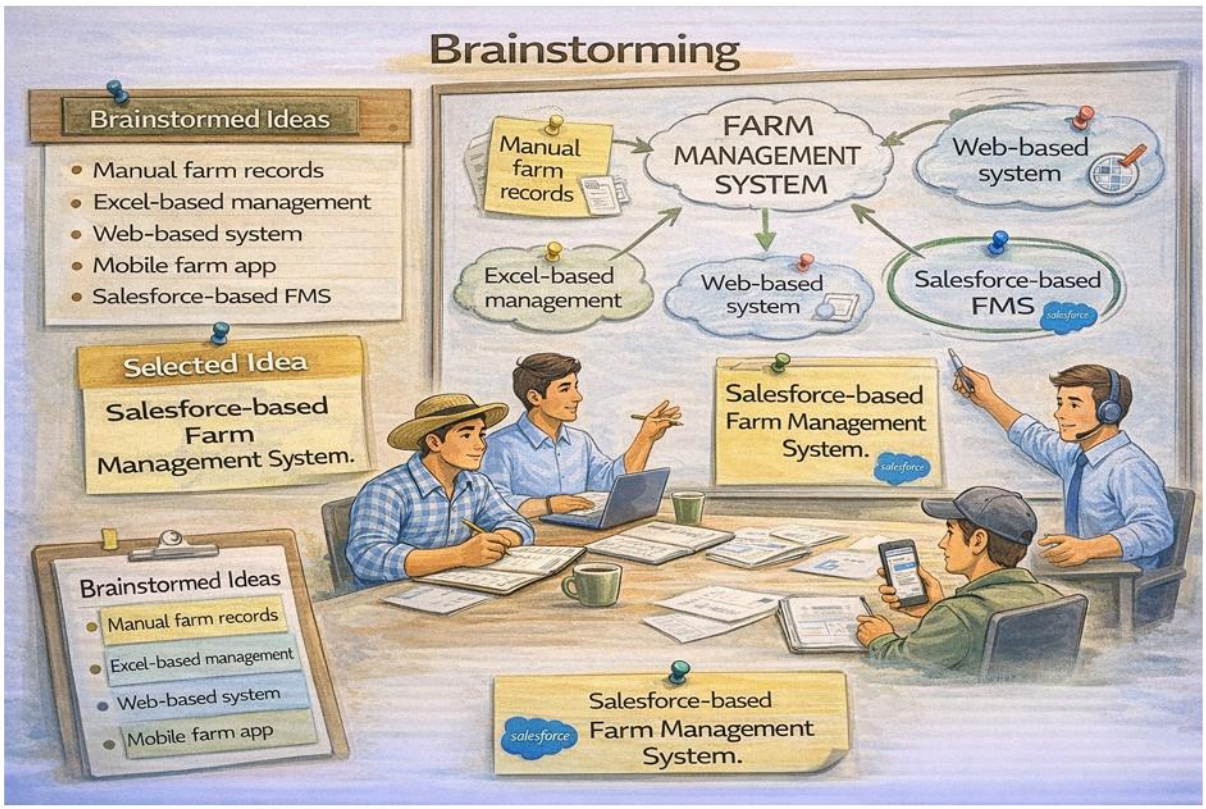
● Manual farm records

● Excel-based management

● Web-based system

● Mobile farm app

● Salesforce-based FMS Selected Idea Salesforce-based Farm Management System.

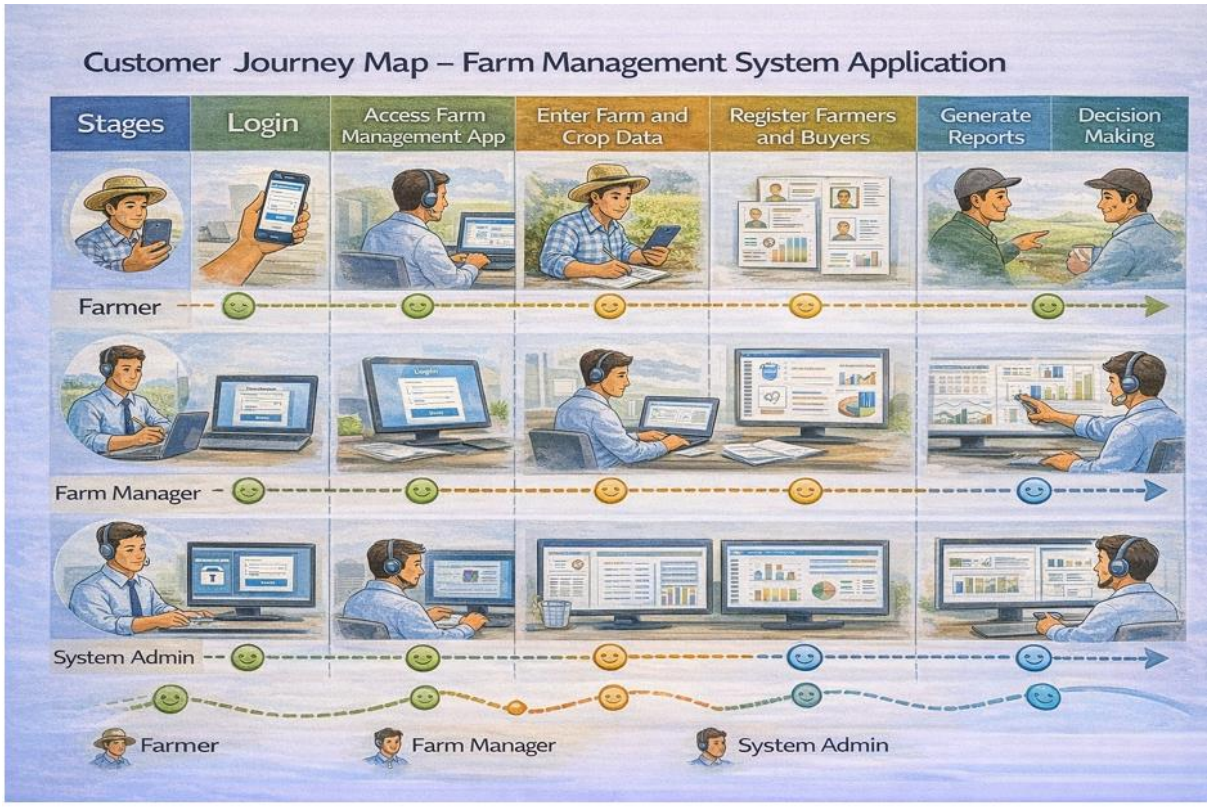


REQUIREMENT ANALYSIS PHASE

Customer Journey Map Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making

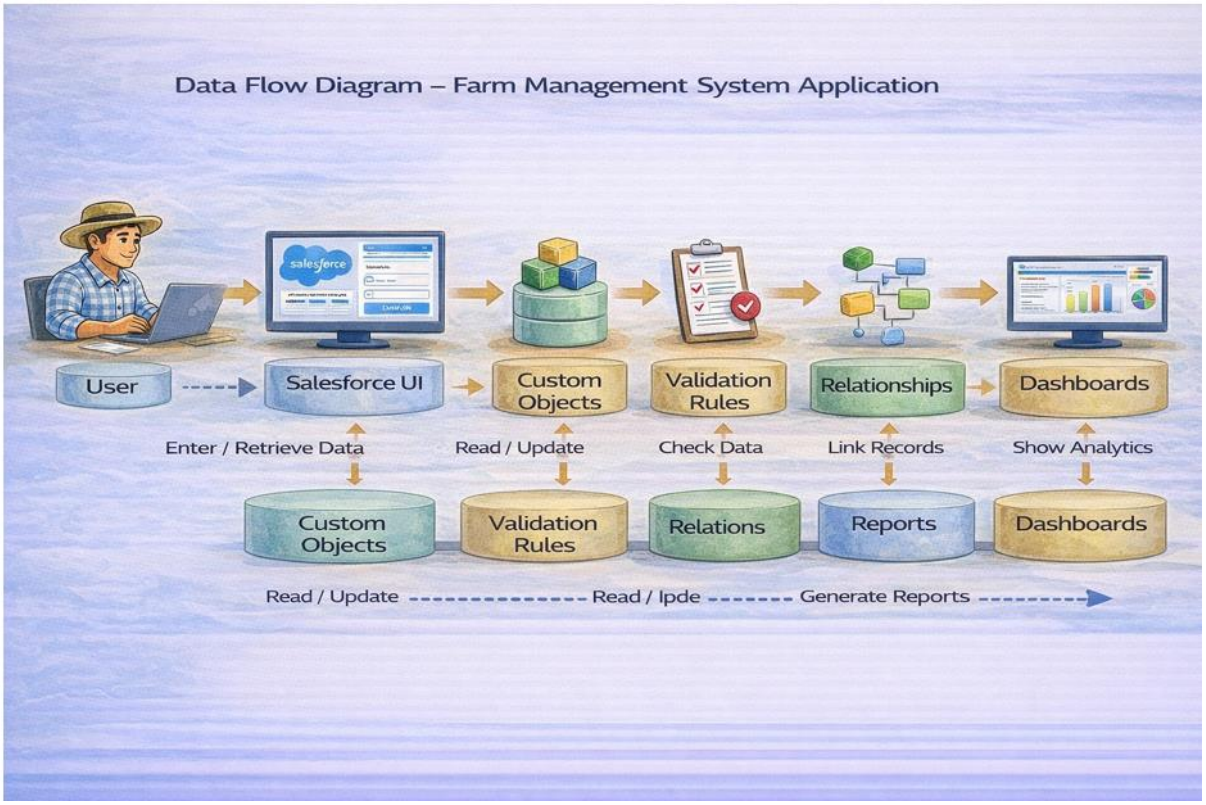
Solution Requirements



Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



Project Development Phase:

Project Description:

Story Points Completed (as on Planned End Date) 20 Sprint Release Date (Actual) 29 Oct 2022 The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

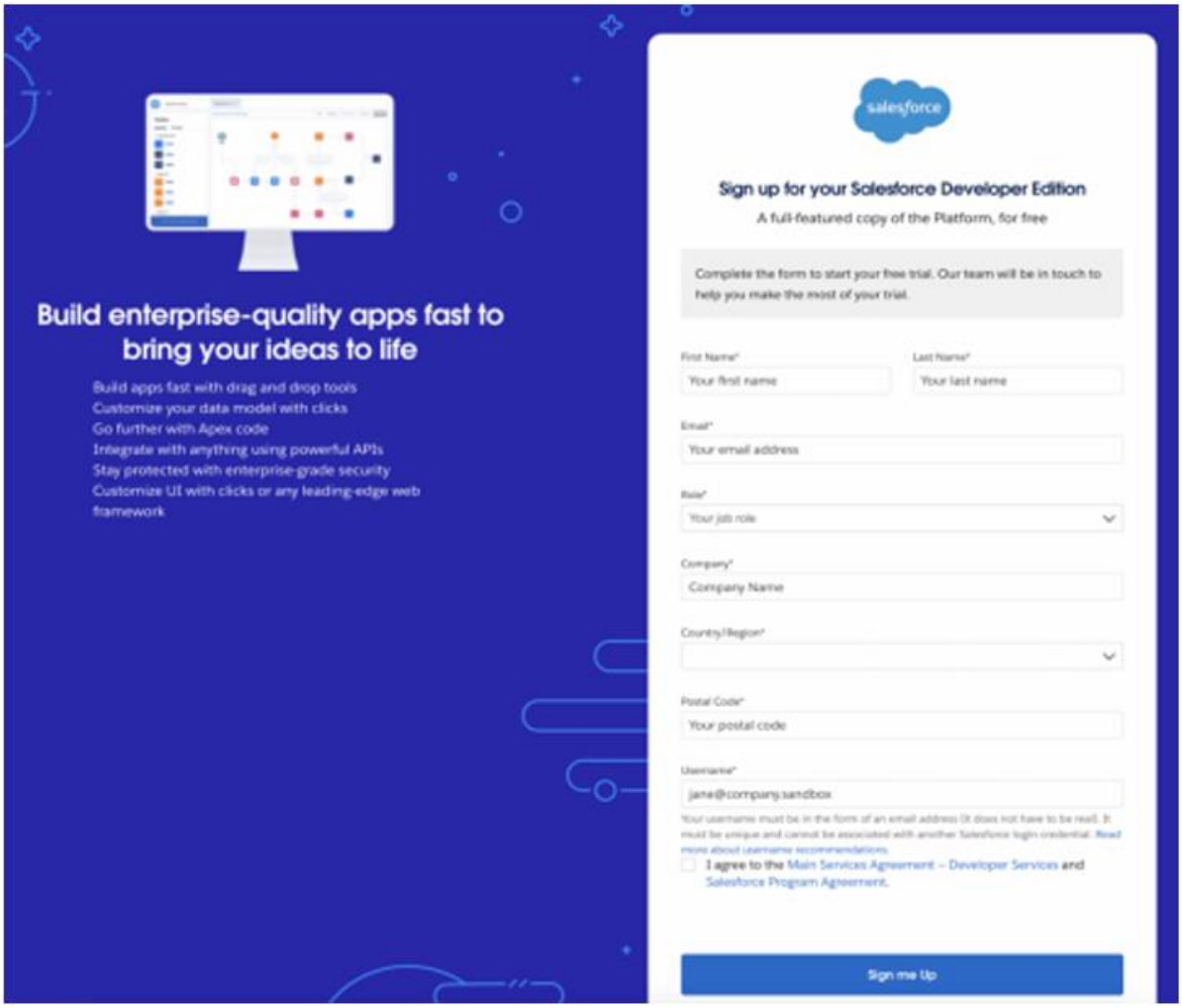
1. Search Developer.salesforce.com/signup

2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.

3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.

4. Click save.

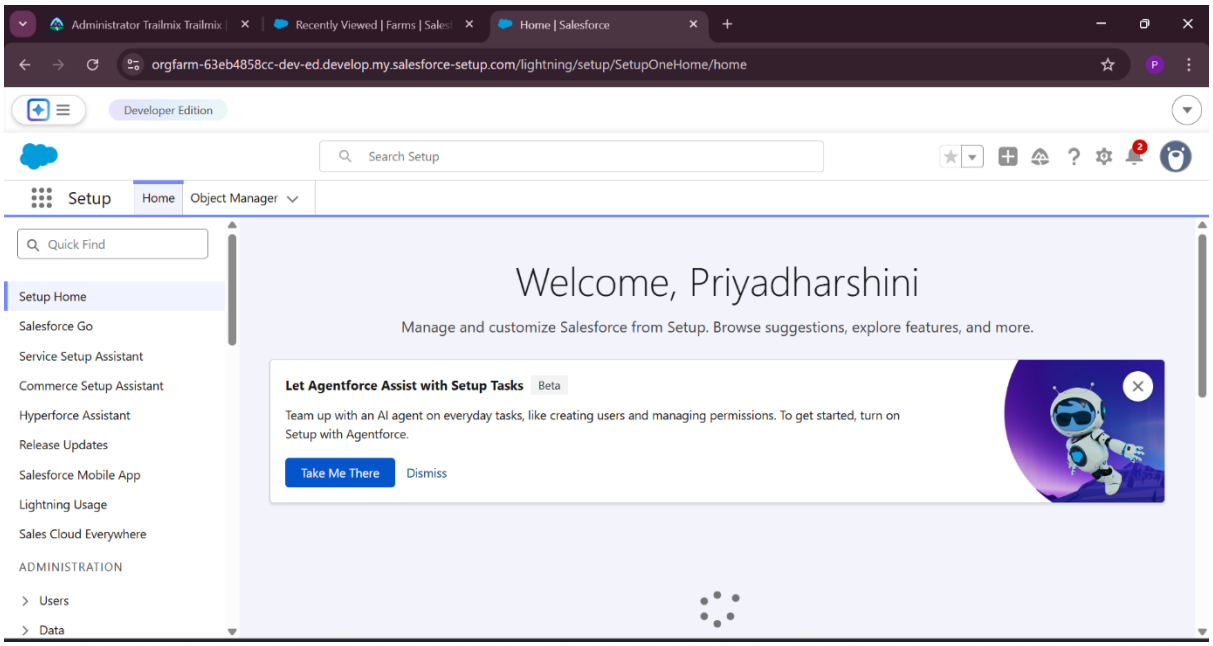
5. Search login.salesforce.com



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows). In This Application We Use 4 Custom Objects:



1.Farm

2.Crop

3.Farmer

4.Buyer

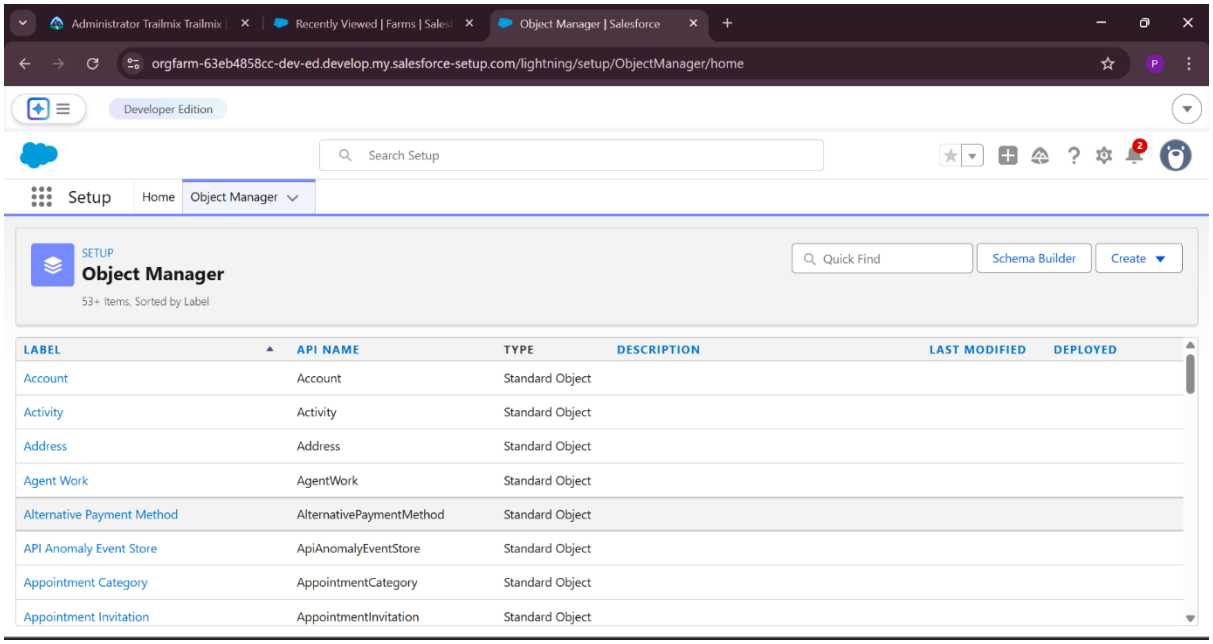
Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.

2. Click create, select custom object.



3. Fill in the label as " Farm ".

4. Fill in the plural label as " Farms".

5. Record name: Farm Name"

6. Select the data type as "Text".

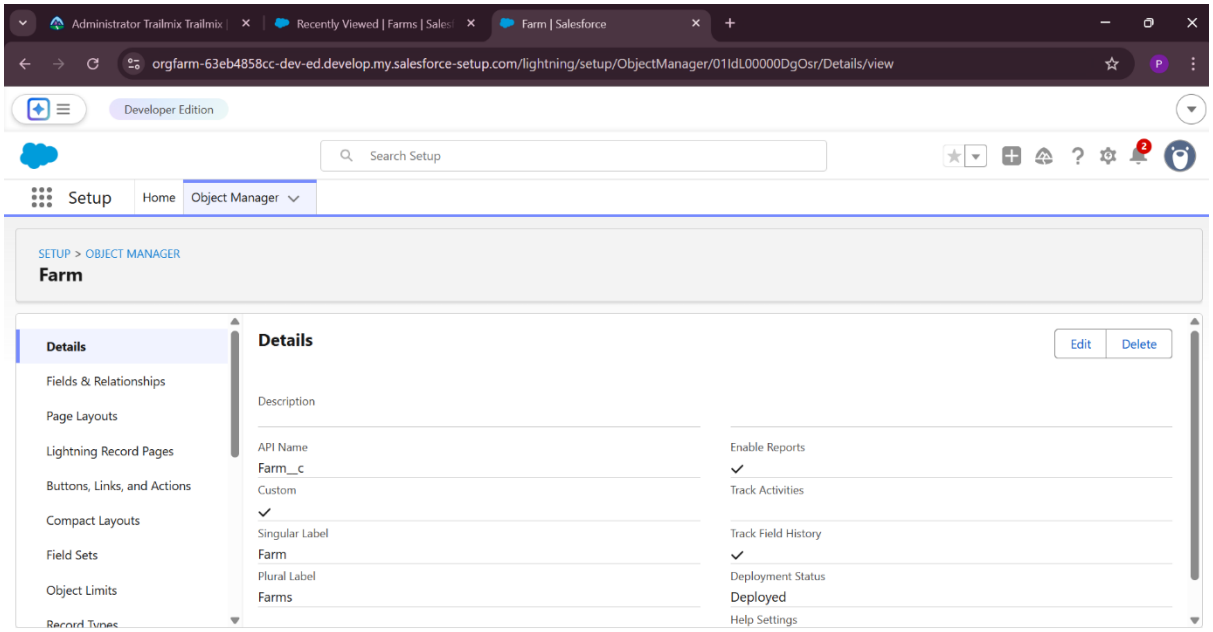
7. In the Optional Features section, select Allow Reports and Track Field History.

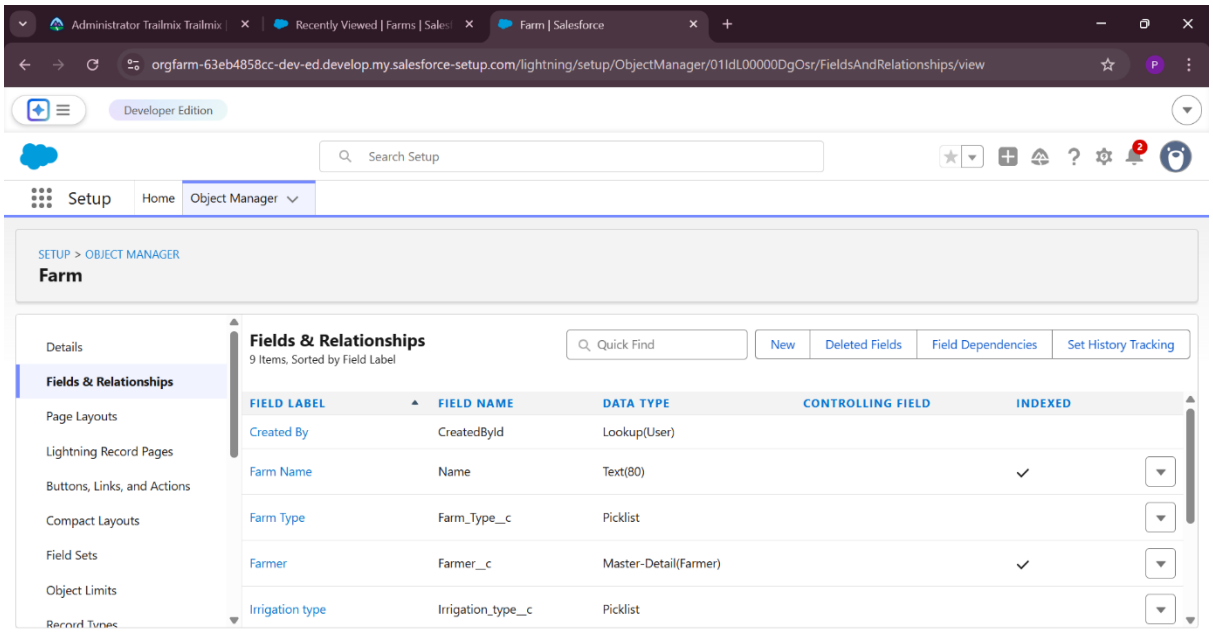
8. In the Deployment Status section, ensure Deployed is selected.

9. In the Search Status section, select Allow Search.

10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

11. Leave everything else as is, and click Save.





Activity 2:

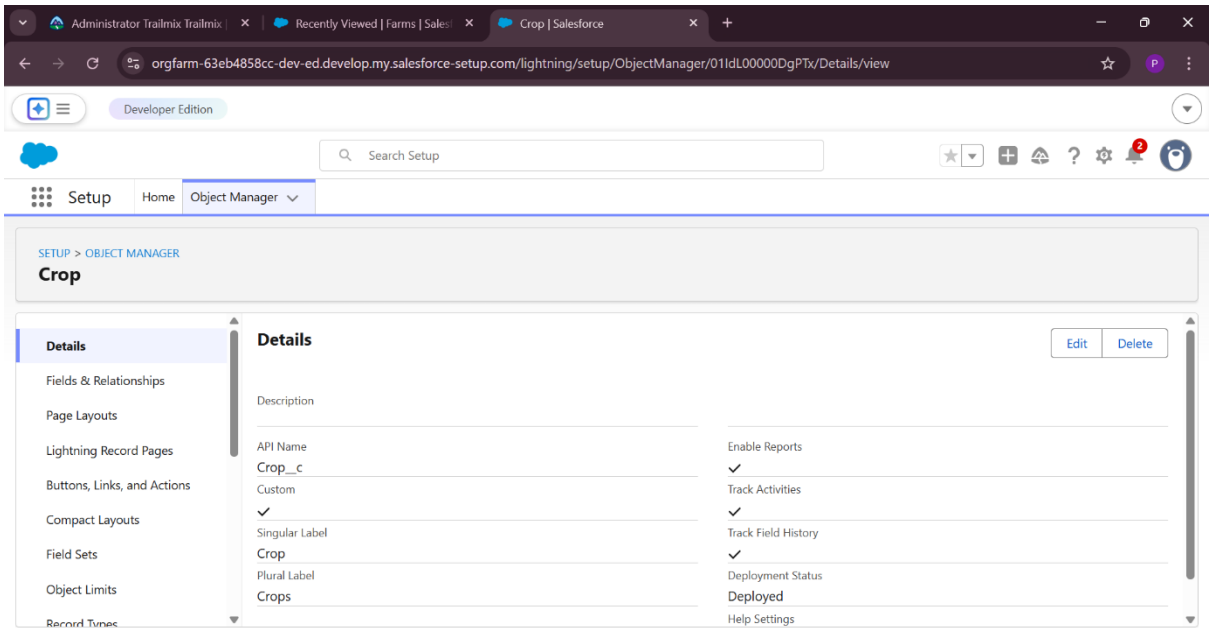
Creation of Crop object:

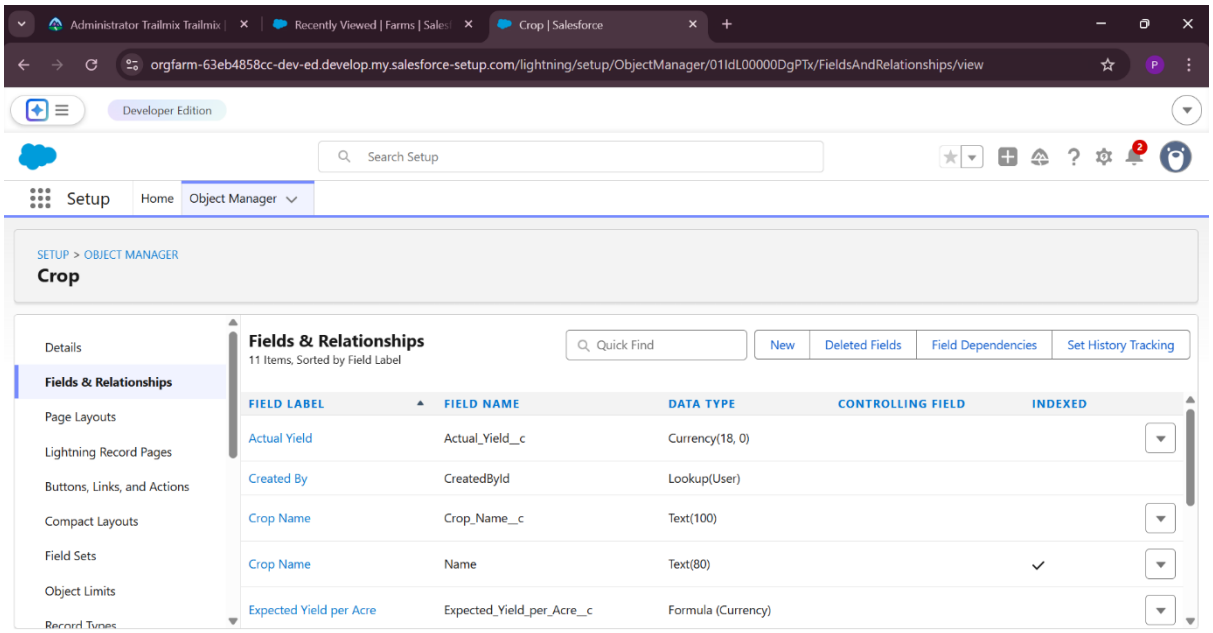
1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object. a. On the Custom Object

Definition page, create the object as follows: b. Label: Crop c. Plural Label: Crops d. Record Name: Crop Name





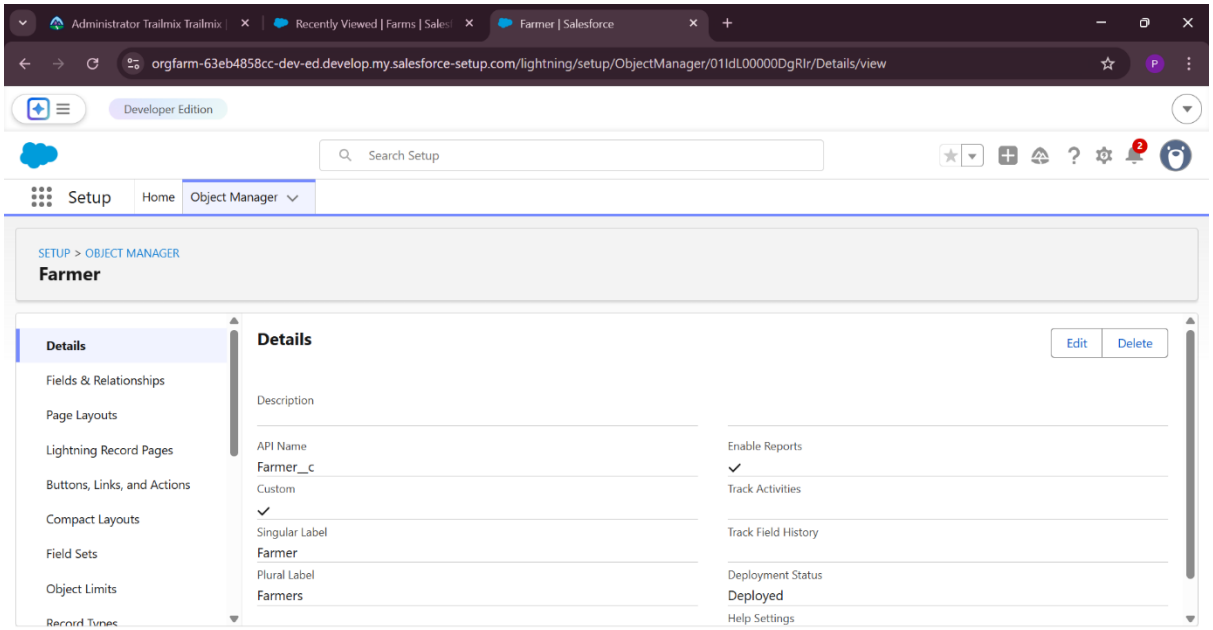
Activity 3:

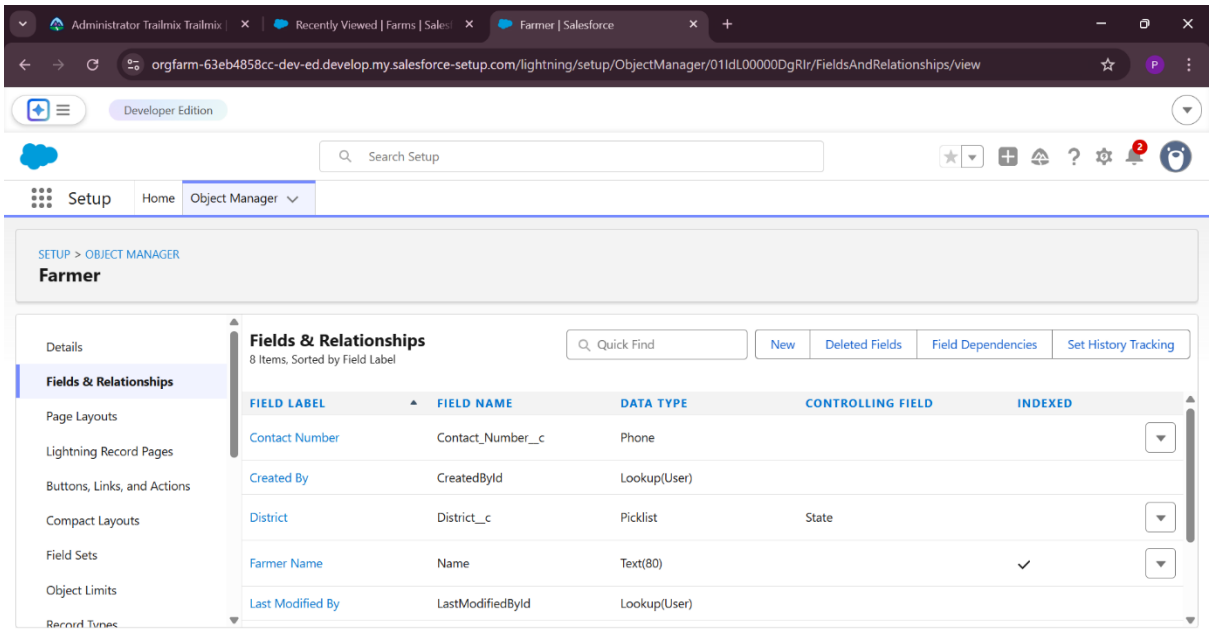
Creation of Farmer object:

1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.





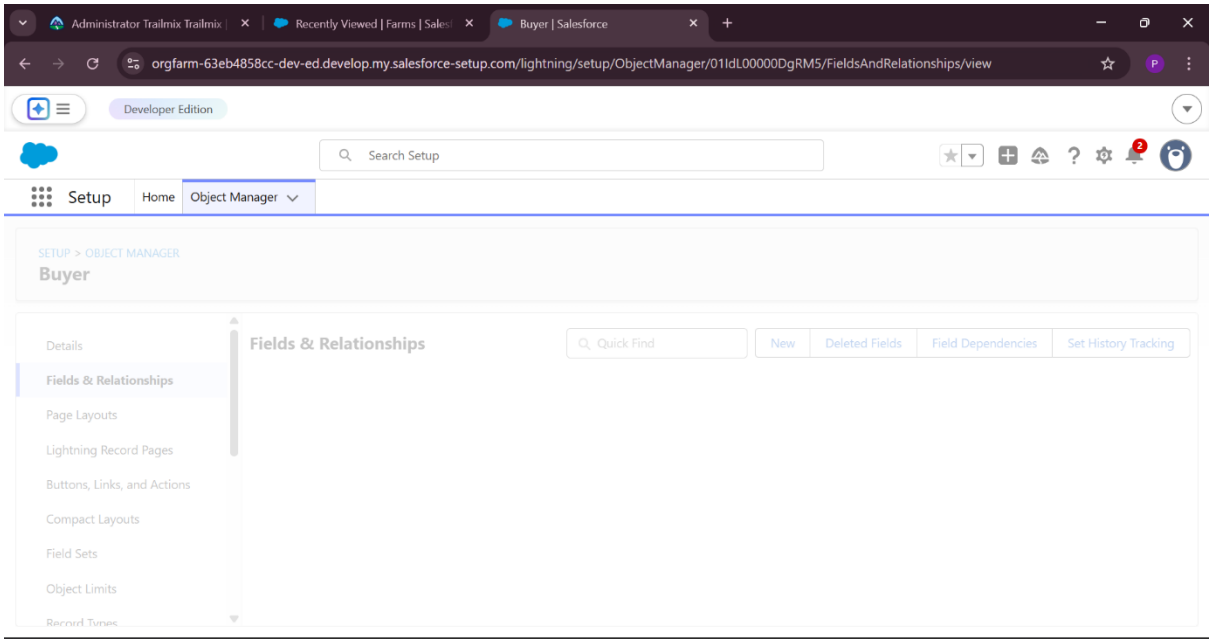
Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.



Milestone 3 - Tab:

Tab Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

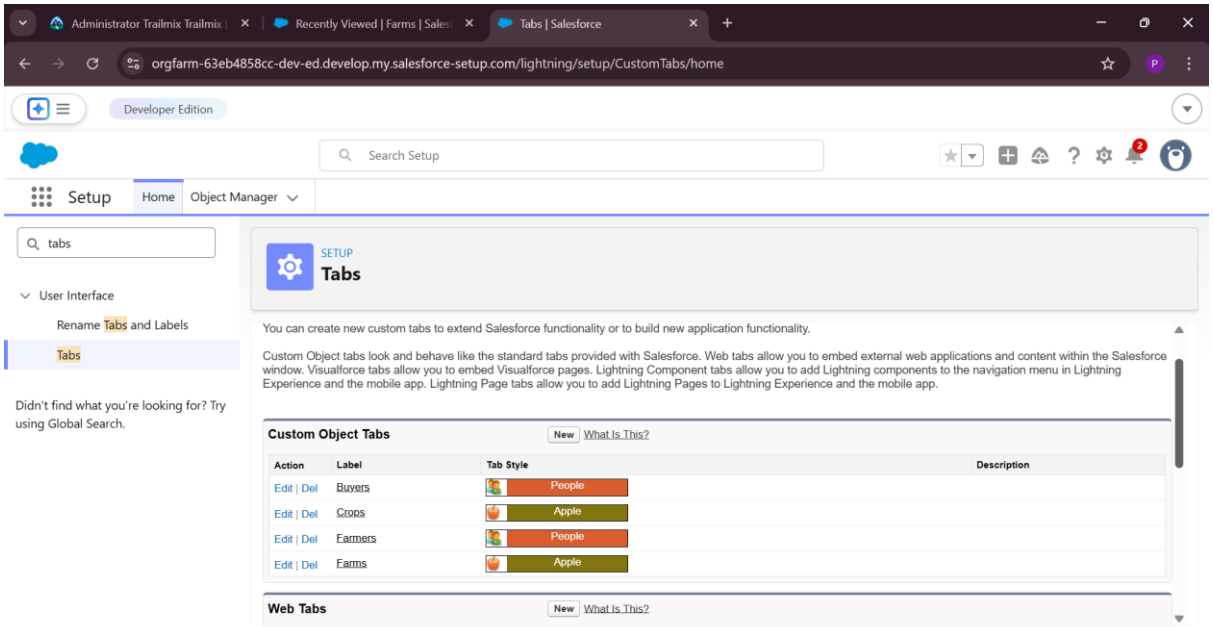
1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Farm.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.



Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

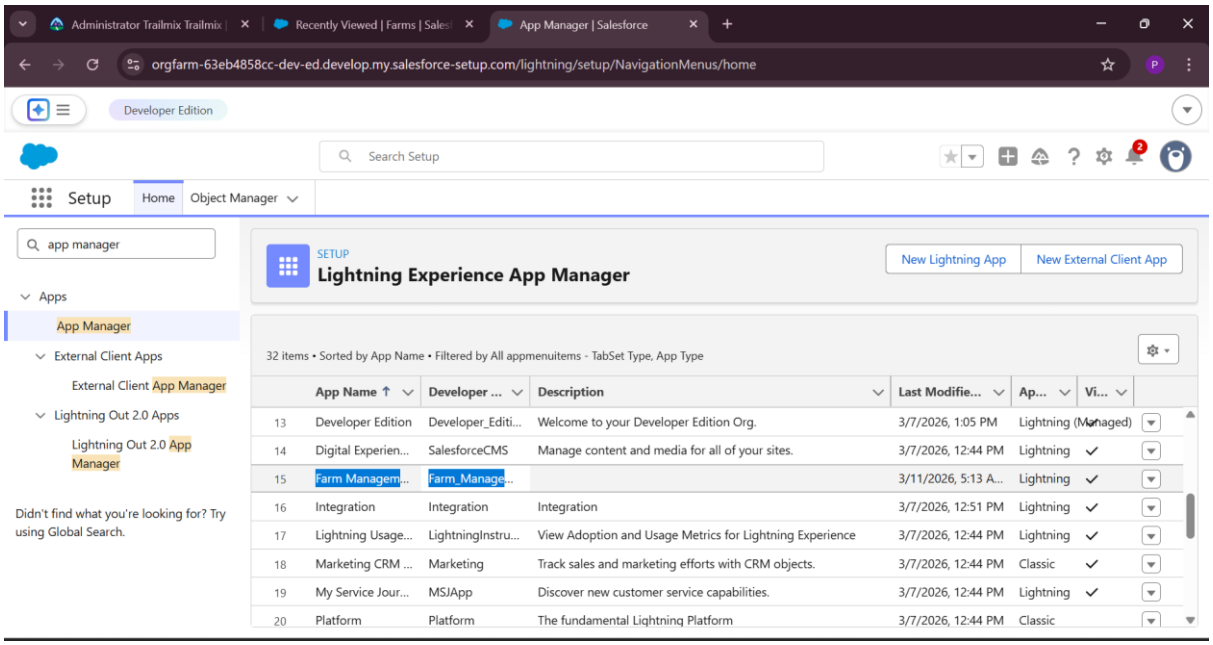
Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together asa unit. It has a name,a logo, and a particular set of tabs. The simplest app usually has just two tabs. Activity 1: Create the Farm Management System app



Milestone 5 - Fields and Relationship:

What are fields? Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Milestone 6- Profile:

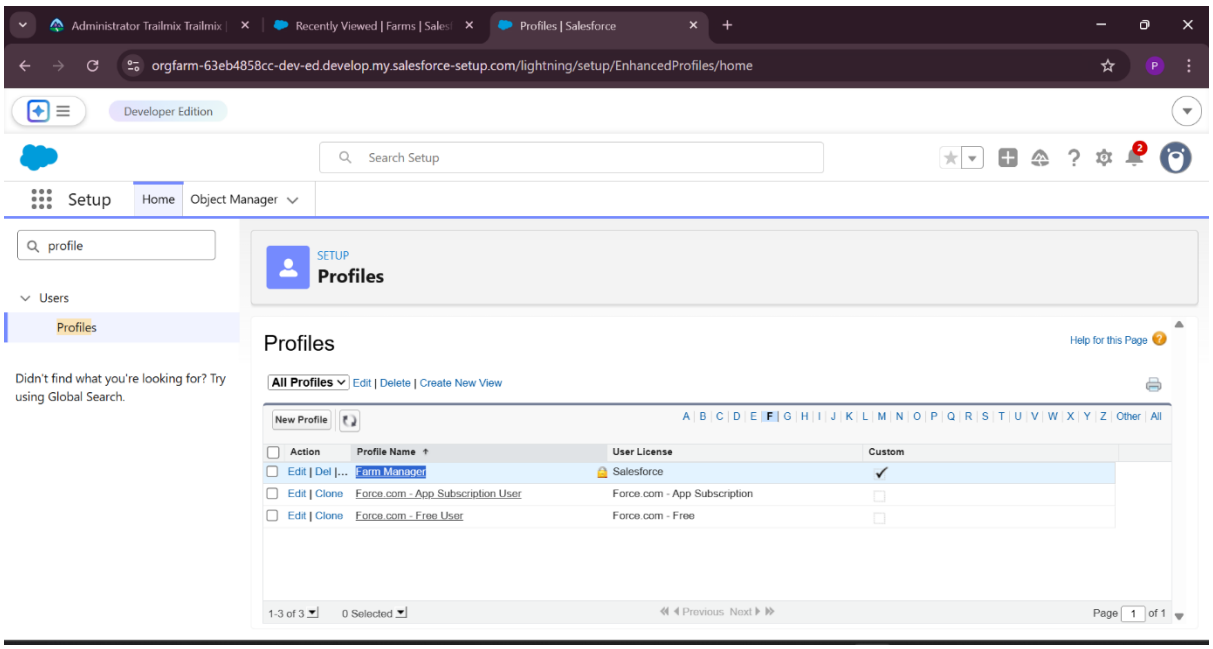
What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges. A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.



Milestone 7-User :

What is a user?

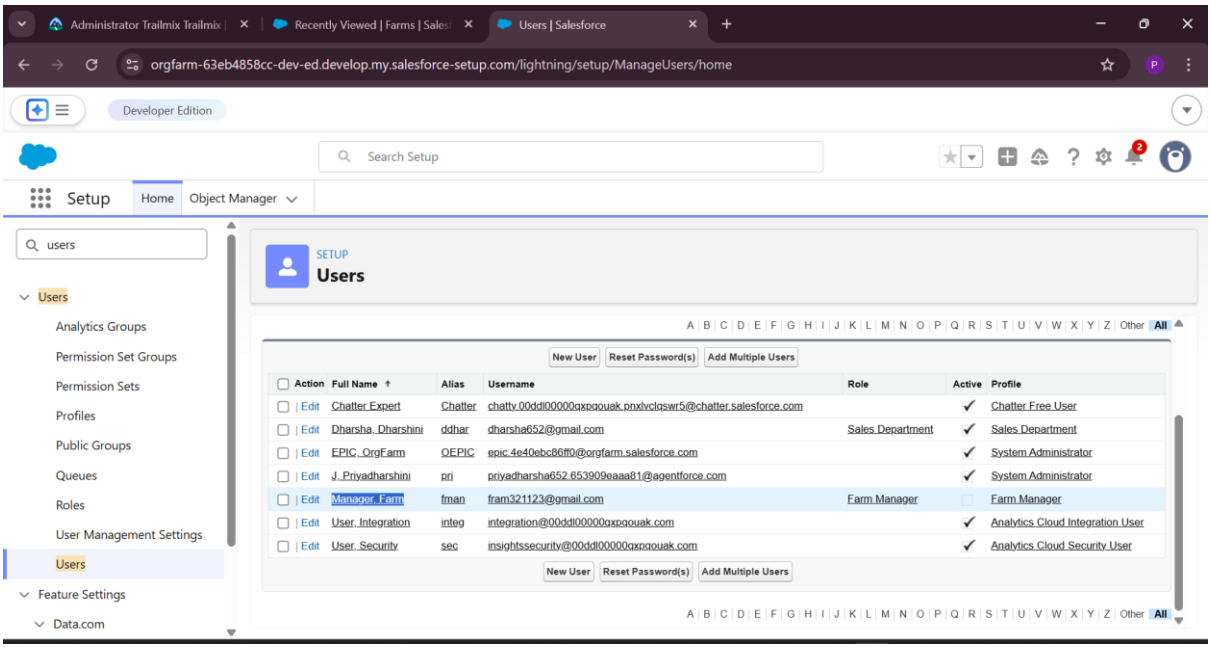
A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users. 2. Click New User.



Milestone 8-Reports:

What are Reports?

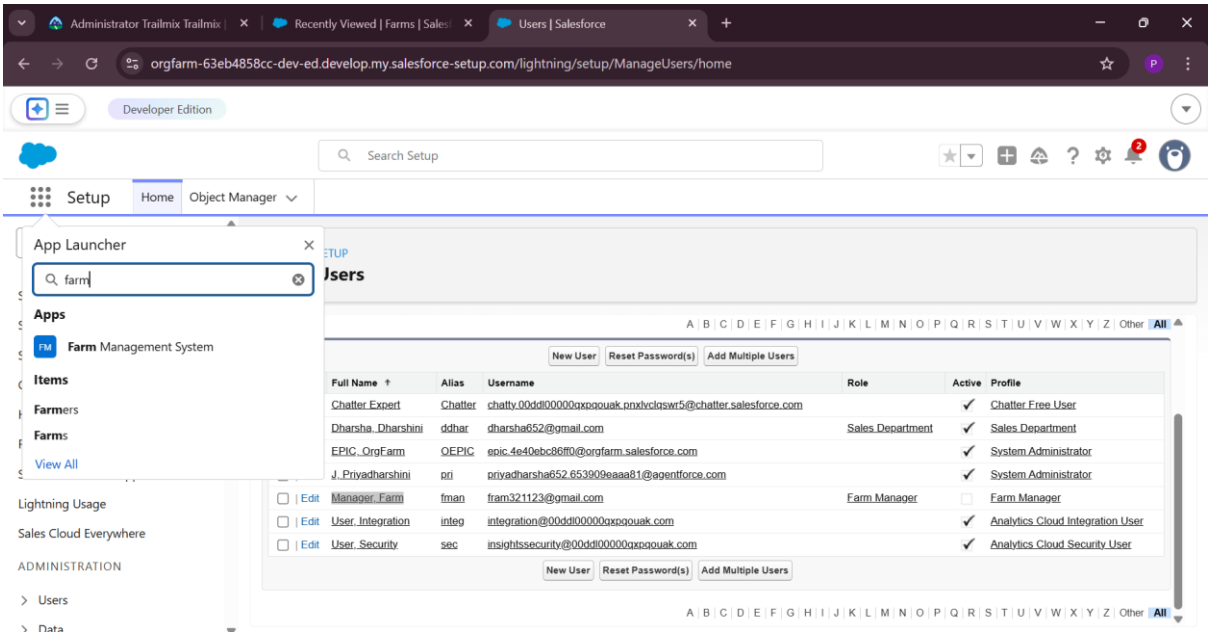
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

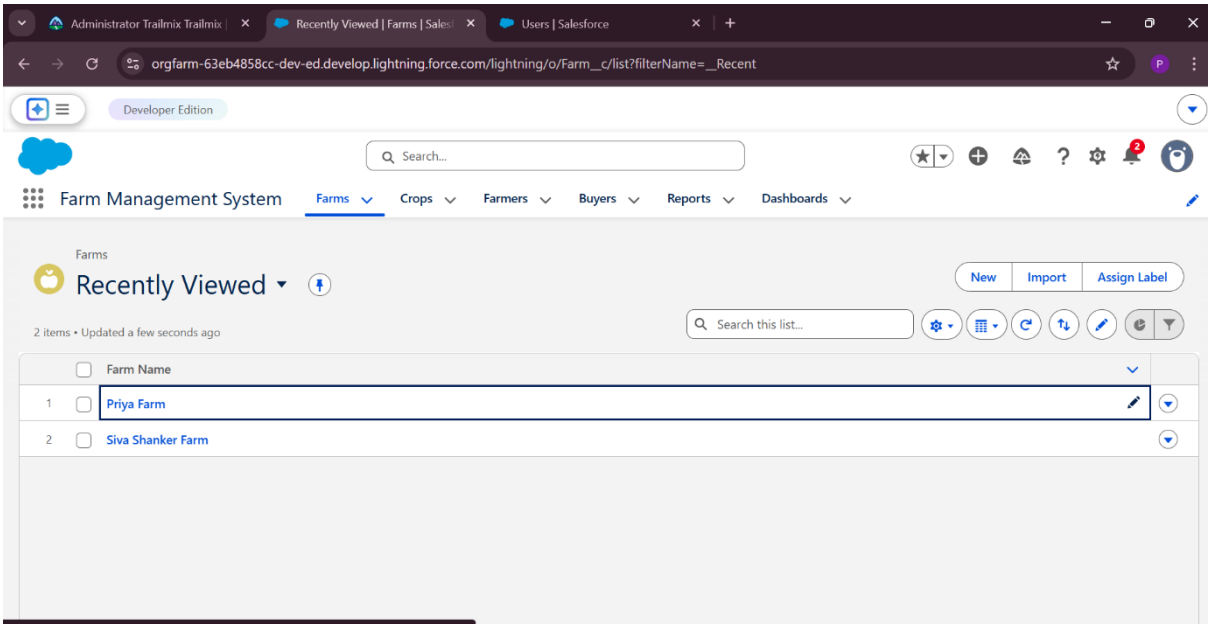
Activity 1:

Create a report: Create a report that displays rating of the account and which has type and account name.

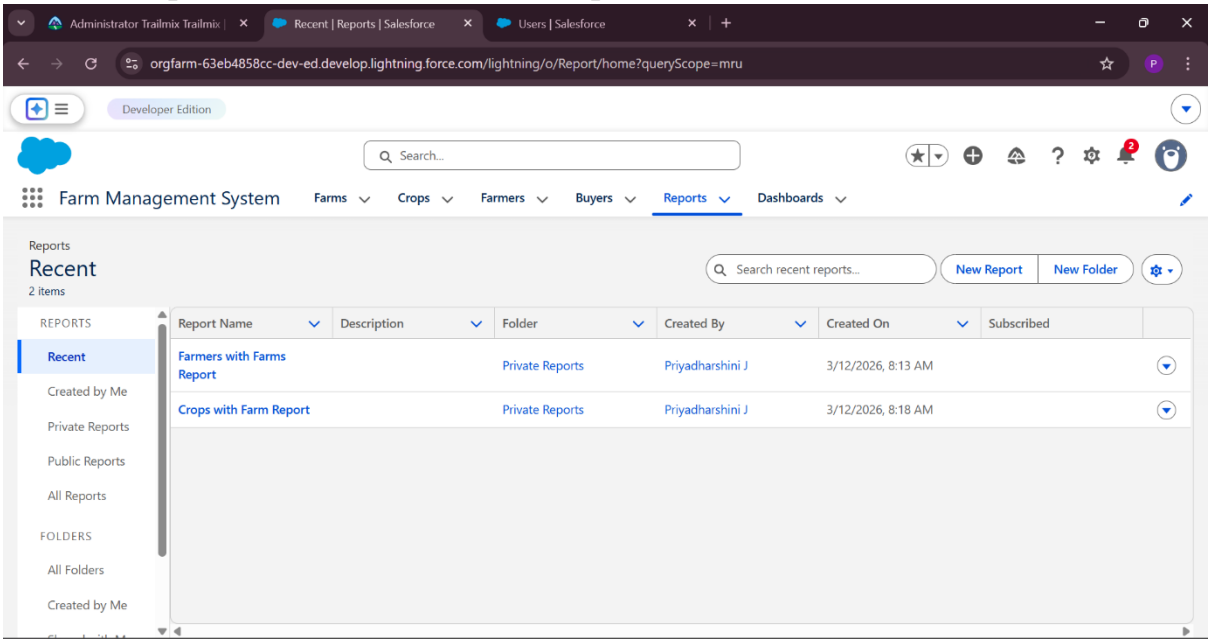
1) Click on app launcher search for reports.

2) Click on the new report and select the category has Farmer with Farms.

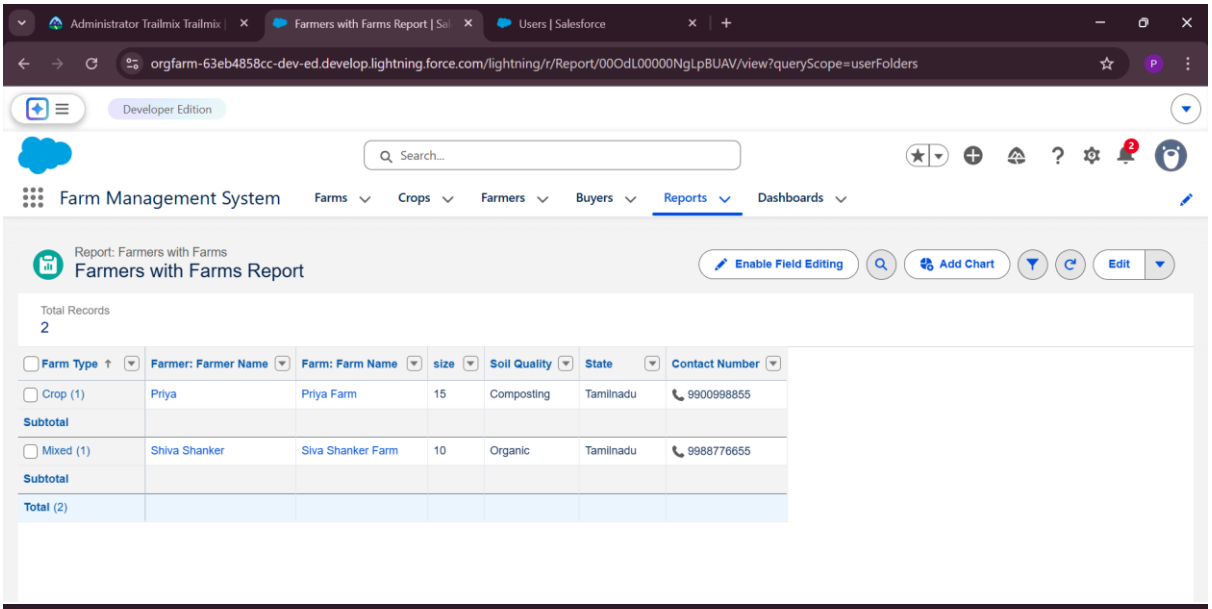


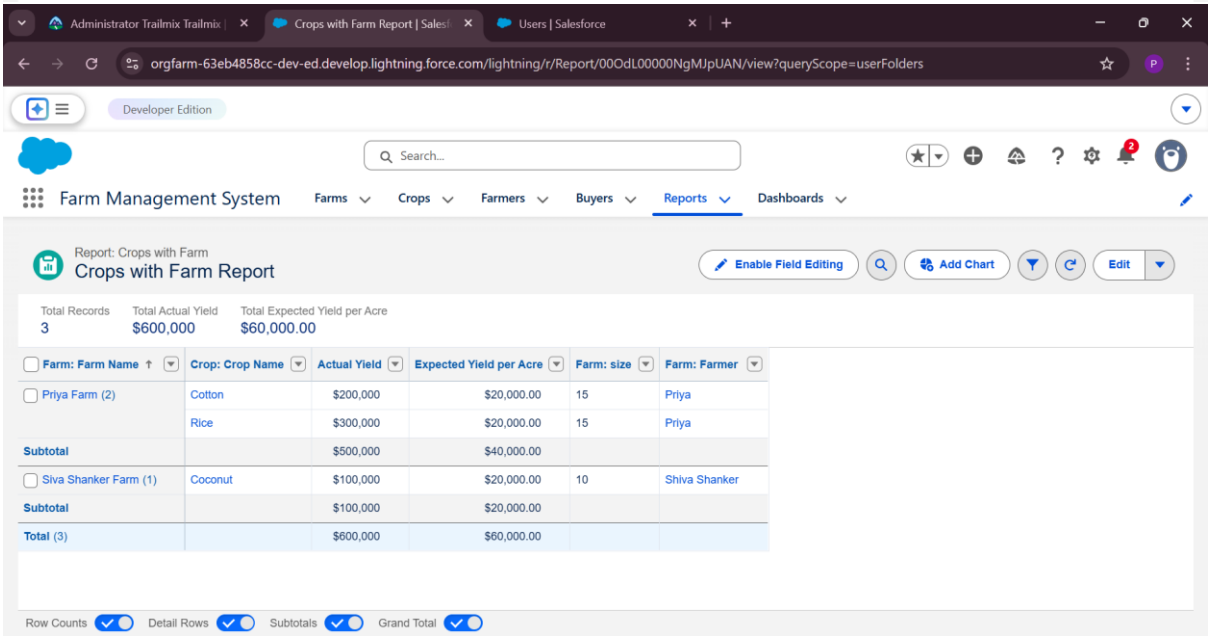


3) In the details section select the option start report.
4) In Outline Click on Column Select Which Field you want to display in the Report.
5) Click Group,Select Which Fields You Want to Group.
6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



Activity 2:
Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.





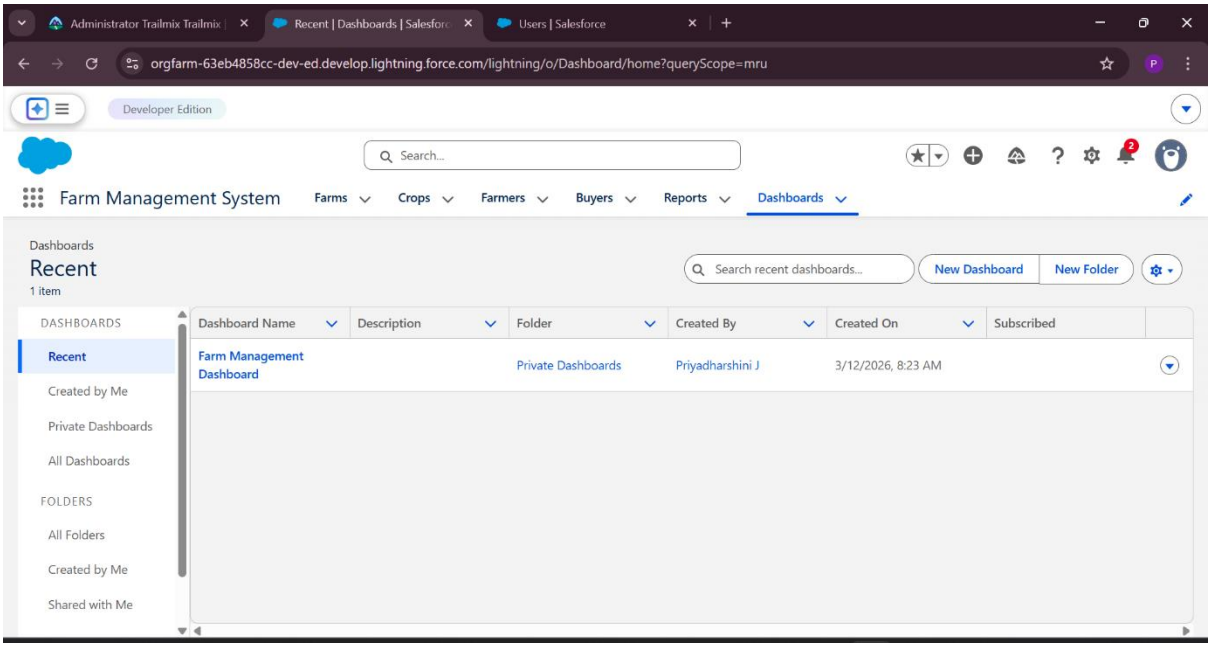
Milestone 9-Dashboard:

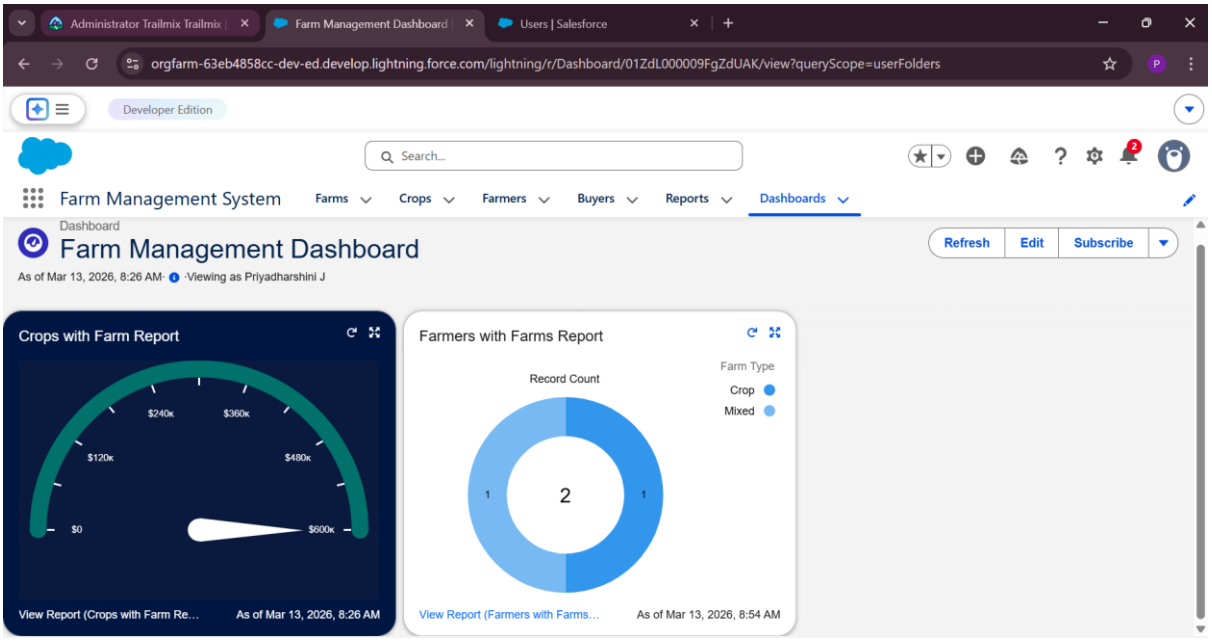
Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1: Create a Dashboard:

1. Click the Dashboard tab.

2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.





FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality.
The following were tested:
● Creation of Farm, Crop, Farmer, and Buyer records
● Validation of input fields
● Relationship and dependency checks
● Profile and role access testing
● Sharing rule verification
● Report and dashboard generation Performance testing ensured smooth system response and stability during multiple user operations.

ADVANTAGES & DISADVANTAGES

Advantages

● Centralized data system
● Reduced paperwork
● Improved accuracy
● Better monitoring
● Real-time analytics

Disadvantages

● Requires internet connection
● Salesforce license dependency
● Limited offline access

Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

FUTURE SCOPE

● Mobile application integration
● AI-based crop prediction
● Weather data integration

- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You